

VLANs — Guided Notes

Unit 2.1.2 | CompTIA Network+ N10-009 Obj. 2.1 | N10-008 2.1

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

A VLAN creates a separate _____ domain on a switch. Devices on different VLANs cannot communicate at Layer 2 even if they are plugged into the _____. VLAN IDs range from _____ to _____. Best practice is to avoid using VLAN _____ for production traffic.

Question 2

An access port belongs to _____ VLAN and connects to _____. A trunk port carries traffic for _____ VLANs simultaneously and typically connects _____.

Question 3

802.1Q tagging inserts a _____ -byte tag into the Ethernet frame on trunk links. The tag contains the _____. When a frame exits an _____ port to an end device, the tag is _____.

Question 4

The native VLAN is sent across a trunk _____. If the native VLAN doesn't match on both ends of a trunk, frames end up _____. Best practice is to change the native VLAN away from _____ and ensure it _____ on both sides.

Question 5

Router-on-a-stick uses _____ on a single router interface — one per VLAN — each configured with the _____ IP for that VLAN. Its limitation is _____. Layer 3 switching uses _____ (SVIs) on the switch itself and routes at _____.

Question 6

A voice VLAN allows one physical port to carry both _____ traffic (untagged, data VLAN) and _____ traffic (tagged, voice VLAN). The IP phone negotiates its VLAN via _____ or _____. QoS

policies on the voice VLAN ensure _____.

Question 7

Match each VLAN misconfiguration to its symptom:

Wrong VLAN assignment → _____

Trunk not formed → _____

VLAN not allowed on trunk → _____

No inter-VLAN routing → _____

Question 8 — Real World Application

REAL WORLD: A new teacher's workstation is connected but can only reach devices in the front office, not classroom resources. Other teachers on the same switch can reach everything. What do you check first and why?

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/2-1-2-vlans/2-1-2-vlans-answer-key.pdf
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